

Boundary adapter qualified; cell-1 film solve remains unresolved

Thermodynamic derivative gate: passed. The separate excess-only adapter passed all required checks across 27 composition cases and 15 boundary families at the frozen 298.15 K condition.

Actual cell-1 film qualification: not achieved. Seven starts produced no admissible numerical candidate. Four were stopped on negative numerical trial profiles and three reached their time limit. No negative trial, timeout or incomplete result was accepted.

1. What caused the earlier derivative failure?

The active path contains two different composition floors: 10^{-12} in the assembled wrapper and 10^{-10} in the native wrapper. Controlled layer-by-layer tests isolate their effect on the nearly solvent-free derivative audit. The fully unfloored native-plus-RK calculation removes the increasing cross-derivative inconsistency while preserving the interior equations.

This is an internal evaluation issue, not evidence of trace NMP or water in the physical fresh RRBO. Its prescribed solvent entries remain exactly zero.

2. Why the new adapter is more than deleted floors

- Exact pinned NIST cCOSMO source commit, binary hash, profile data and RK parameter lineage were checked. No library was replaced or rebuilt.
- The combinatorial expression is already algebraically ratio-reduced and finite at zero component concentration when mixture geometry is positive.
- The residual uses the absent species' own pure profile. Mixture profile weights vanish at zero concentration without deleting the absent species' infinite-dilution excess activity.

- Positive-kernel segment equations provide an exact-equation existence and local implicit-function regularity argument, with actual profile, kernel and fixed-point checks against the pinned binary.
- The original additive RK polynomial is retained. Ideal $\ln(x)$ is separate; finite excess activity does not make ideal $\ln(0)$ finite.

Source correspondence plus ABI checks is not reproducible-build certification. Finite-difference checks assess the numerical evaluator; the analytic argument applies to the exact segment equations.

3. Independent qualification results

Check	Worst observed	Unchanged limit
Gibbs-Duhem residual	2.255×10^{-7}	2×10^{-6}
Tangent-integrability residual	1.256×10^{-6}	2×10^{-5}
Relative derivative-refinement drift	1.556×10^{-7}	2×10^{-3}
Interior chemical-potential agreement	1.827×10^{-14}	2×10^{-10}
Terminal-three zero-limit agreement	2.342×10^{-12}	2×10^{-10}

Additional checks passed: native combinatorial agreement, independent residual reconstruction, nonnegative normalized pure/mixture profiles, positive symmetric kernel, actual mixture and pure segment fixed-point residuals, invalid-input refusal and unchanged literal-zero vectors.

The active segment iteration tolerance remains 10^{-8} . Its damped stopping rule implies a fixed-point residual bound $2t+t^2$; the worst measured 1.381×10^{-8} is below $2.00000001 \times 10^{-8}$. That bound is derived from the existing rule, not a loosened tolerance.

4. The initially failed approach point remains failed

At the pure-water corner, the equal-component approach with $\varepsilon=10^{-12}$ had a 2.312×10^{-10} value difference, slightly above the 2×10^{-10} limit. The entire original failed report and curve remain preserved.

One additional mathematical refinement, $\varepsilon=10^{-18}$, supplied three consecutive finer passing points: $\varepsilon=10^{-14}$, 10^{-16} , 10^{-18} , with differences 2.342×10^{-12} , 2.998×10^{-14} , 5.329×10^{-15} . The evaluated terminal window moved; the limit, model, cases and physical feed did not. These approach compositions are numerical limit tests, not feed data.

5. Actual saved cell-1 attempts

The frozen bulk vectors, Job-A conductances, phase orientation, temperature and archived state hashes were retained. Total molar flux was free: an initial zero guess was not an $n=0$ constraint. No archived flux was accepted as a new finite-film result.

Start	Stop reason	Elapsed
ARCHIVED_INTERFACES_NEW_FREE_FLUX	Numerical time budget	30.00 s
ORIENTED_PHASE_TEMPLATE	Negative numerical trial profile — rejected	17.27 s
FROZEN_BULK_BOUNDARIES	Numerical time budget	30.00 s
ORIENTED_TEMPLATE_BULK_C	Negative numerical trial profile — rejected	0.20 s
ORIENTED_BULK_D_TEMPLATE	Numerical time budget	30.00 s
ORIENTED_PHASE_TEMPLATE_PERTURB_PLUS	Negative numerical trial profile — rejected	17.33 s
ORIENTED_PHASE_TEMPLATE_PERTURB_MINUS	Negative numerical trial profile — rejected	17.28 s

The initial uncached attempt was time-limited. A bounded exact-state cache then avoided repeated identical evaluations; its outputs were verified bitwise equal on the saved states, including cache misses and hits. No composition was rounded and no thermodynamic value interpolated.

Operational timeouts can be wrapped by native/RK exception handlers. The instrumented attempt records their causes and labels them as time budgets, not undefined thermodynamics. Negative *intermediate numerical trials* do not prove that all possible physical solutions have negative profiles.

6. What is—and is not—established

The original derivative blocker has been resolved for this isolated, source-bound adapter. The film problem has not been solved or accepted. No candidate exists on which to complete independent profile refinement, Jacobian-rank, reproduction or interface stability/TPD acceptance. These checks are not marked passed or bypassed.

Next numerical requirement: keep trial updates inside the admissible composition domain and control derivative-evaluation cost, then repeat the single-cell attempt under the same physics and gates. A further attempt is not a column run, and no process-infeasibility conclusion follows from this result.

No new accepted flux, active height, physical-stage count, efficiency, final RPM or final diameter follows. Production Jobs A/B/C remain unchanged.

7. Reproducibility

Commands and caveats are in `research/finite_film/README.md`. Machine-readable evidence:

- `research-results/finite-film-boundary-qualification.json`
64d54cc1adeb0bd39748de220f28a5e1b9f5f543a602b6d878793adcadcb2ee5
- `research-results/finite-film-boundary-qualification-initial-failed.json`
65911fc1e9dbf12c384fa3f5f9476aeb294abaeb53e143a03794ae6f5ad61f04
- `research-results/finite-film-boundary-native-segment-proof.json`
9daae07b6c9d8aed73e7466da40c3636f6070d5dff0115262069a466b1460713

- `research-results/finite-film-cell1-attempt.json`
`0502f251dd0d60e67ffca0d359f426329fab1f71f6a63cc64658139272c458e7`

Overall status: DERIVATIVE GATE PASSED / NUMERICAL CELL HOLD. Effective-resistance mobility remains an unvalidated transport assumption.